

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	follow procedures and record observations using suitable scaled instruments and units
KEY IDEA	Choose an instrument and unit that match the quantity. Read the scale at eye level, record units every time and distinguish a measurement from a description.
WORKED MODEL	Use the inquiry cycle: question -> prediction -> safe method -> observations/data -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN

[] use the inquiry skill [] record evidence clearly [] evaluate quality [] explain a next step

A. BUILD THE SCIENCE IDEA

1. Match thermometer, ruler, measuring cylinder and stopwatch to quantity and unit.

2. Read four described scales by identifying the value of each small division.

3. Classify records as qualitative observations or quantitative measurements.

B. APPLY AND EXPLAIN

4. Correct this record: water = 25; leaf = greenish; time = about 3.

5. Explain why eye position and starting at zero matter.

6. Prepare a table with headings, units, trials and space for observations.

C. INDEPENDENT CHECK

7. Record three repeated measures and describe whether they are consistent.

8. Decide whether to repeat an unusual result; justify without deleting evidence automatically.

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D. APPLY SCIENCE IN CONTEXT

9. Scenario: use scaled instruments to measure how surface type affects a toy car's travel distance

10. Select equipment, scale, unit and measurement intervals.

11. Write a procedure for recording both numerical data and useful observations.

E. REASON, EVALUATE AND TRANSFER

12. Diagnose three errors in a table with mixed units and missing headings.

13. Improve measurement precision without claiming the instrument is more accurate than its scale.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can follow procedures and record observations using suitable scaled instruments and units.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Temperature/C; length/mm or cm; volume/mL; time/s or min.
2. Divide labelled interval by number of equal spaces.
3. Qualitative uses words; quantitative uses numbers and units.
4. Add C, make description specific, measure time with a stated unit.
5. Parallax and an incorrect zero shift readings.
6. Table needs title, variables, units and trials.
7. Compare range and pattern rather than demanding identical values.
8. Retain, check method and repeat where possible.
- 9-11. Choices must match the scenario and create traceable records.
12. Standardise units and add precise headings.
13. Use a finer suitable instrument and report only justified precision.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

More decimal places do not automatically make a measurement more accurate.

Do not replace an observation with a guess when an instrument can measure it.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.